

4.13.16

EE25BTECH11011 - Navya

Question:

The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0$ are perpendicular to a common line for

- 1) exactly one value of p
- 2) exactly two values of p
- 3) more than two values of p
- 4) no value of p

Solution:

Given lines are,

$$p(p^2 + 1)x - y + q = 0 \quad (1)$$

$$(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0 \quad (2)$$

The normal vectors of the lines are

$$\mathbf{n}_1 = \begin{pmatrix} p(p^2 + 1) \\ -1 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} (p^2 + 1)^2 \\ p^2 + 1 \end{pmatrix} \quad (3)$$

If both lines are perpendicular to a common line with normal vector $\mathbf{m}$, then

$$\mathbf{n}_1^T \mathbf{m} = 0, \quad (4)$$

$$\mathbf{n}_2^T \mathbf{m} = 0. \quad (5)$$

Hence, the two equations can be written together as

$$\begin{pmatrix} p(p^2 + 1) & -1 \\ (p^2 + 1)^2 & p^2 + 1 \end{pmatrix} \mathbf{m} = \mathbf{0} \quad (6)$$

Writing the system as an augmented matrix:

$$\left[\begin{array}{cc|c} p(p^2 + 1) & -1 & 0 \\ (p^2 + 1)^2 & p^2 + 1 & 0 \end{array} \right] \quad (7)$$

Perform row operations to reduce to RREF:

$$\left[\begin{array}{cc|c} p(p^2 + 1) & -1 & 0 \\ (p^2 + 1)^2 & p^2 + 1 & 0 \end{array} \right] \xrightarrow[R_1 \rightarrow \frac{1}{p(p^2 + 1)} R_1]{} \left[\begin{array}{cc|c} 1 & -\frac{1}{p(p^2 + 1)} & 0 \\ (p^2 + 1)^2 & p^2 + 1 & 0 \end{array} \right] \quad (8)$$

$$\xrightarrow{R_2 \rightarrow R_2 - (p^2 + 1)^2 R_1} \left[\begin{array}{cc|c} 1 & -\frac{1}{p(p^2 + 1)} & 0 \\ 0 & (p^2 + 1) + \frac{(p^2 + 1)^2}{p(p^2 + 1)} & 0 \end{array} \right] \quad (9)$$

$$= \left[\begin{array}{cc|c} 1 & -\frac{1}{p(p^2+1)} & 0 \\ 0 & (p^2+1)\left(1+\frac{1}{p}\right) & 0 \end{array} \right] \quad (10)$$

$$\xrightarrow{R_2 \rightarrow \frac{1}{(p^2+1)\frac{p+1}{p}} R_2} \left[\begin{array}{cc|c} 1 & -\frac{1}{p(p^2+1)} & 0 \\ 0 & 1 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 + \frac{1}{p(p^2+1)} R_2} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \quad (11)$$

For a non-trivial solution $\mathbf{m} \neq \mathbf{0}$, the system must be **rank deficient**, meaning the second row before normalization must be zero:

$$(p^2+1)\frac{p+1}{p} = 0. \quad (12)$$

Solve for p :

$$p^2+1=0 \implies p = \pm i \quad (\text{non-real, discard}), \quad (13)$$

or

$$p+1=0 \implies p = -1. \quad (14)$$

Check special case $p = 0$:

If $p = 0$, the matrix becomes

$$\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \quad (15)$$

which has full rank, so only the trivial solution exists.

The only real value of p such that both lines are perpendicular to a common line is

$$\boxed{p = -1} \quad (16)$$

